Idea generation & development

National 5 · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe the key stages of brainstorming
- ☐ describe the key stages of morphological analysis
- ☐ explain why idea-generation techniques are used
- ☐ explain exploration, synthesis and refinement of ideas

I DO — Worked example

Read how a full-mark answer is built

A design team needs ideas for a new bike light.

1 Describe three key stages of brainstorming. (3 marks)

Full-mark answer each ✓ is one mark

The problem is explained clearly to a small group so everyone knows the purpose ✓. The group suggests as many ideas as possible within a time limit, and every idea is recorded ✓. No idea is criticised; members build on each other's ideas before the best are selected ✓.

Why marks are not awarded

"You think of lots of ideas and write them down."

0–1 mark: "think of lots of ideas" was given 0; only recording ideas could score.

WE DO — Guided practice

Use the starters and word bank to help you

2 Describe how morphological analysis would be used to generate ideas for a desk tidy. (3 marks)

Sentence starters

First, …
Next, … using a …
Then …

Word bank

parameters · material · sketch · identify · important · number · headings · features

Tip: 3 marks = 3 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which is a rule of brainstorming?
☐ Only feasible ideas are written down ☐ Every idea is recorded without criticism ☐ Ideas are judged as soon as they appear ☐ It must be done alone to avoid copying

b) How does morphological analysis improve the breadth of ideas?
☐ It creates many unique combinations of variables ☐ It quickly removes the weakest ideas early on ☐ It tests each idea with users before sketching ☐ It records every idea as an annotated sketch

c) Combining the best features of several initial ideas is called...
☐ Synthesis ☐ Refinement ☐ Evaluation ☐ Analysis

4 Key terms: write the letter of the matching description next to each term.

1. **Morphological analysis**	____	**A** A matrix of design variables and options; combine one option from each to generate many ideas.
2. **Brainstorming rule**	____	**B** Record every idea and don't criticise; build on each other's ideas within a time limit.
3. **Synthesis**	____	**C** Combining the best features of several ideas into one stronger idea.

5 Explain what is meant by ‘Refinement’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?